zunair zafar

Date of birth: 07/01/2001

Nationality: Pakistani

Gender: Male

CONTACT

zakariya town street # 32 , House # 06, Multan 60000 Multan, Pakistan **(Home)**

zunairzafffar@gmail.com

(+92) 3219905434

https://github.com/zunairzafar

ID: Zunair_Zafar **(WeChat)**

WORK EXPERIENCE

Alethea A.I. Multan, Pakistan

Dataset Integration Engineer

09/06/2021 – 13/09/2023

- Processed and cleaned 50,000+ audio samples using automated pipelines, reducing data noise and improving downstream model training accuracy.
- Performed exploratory data analysis (EDA) on multimodal datasets (audio, text, metadata), surfacing key quality issues that reduced data rejection rates by 35%
- Built automated audio preprocessing scripts (noise reduction, normalization, segmentation) in Python, cutting manual cleaning time by 60% across the data team
- Collaborated cross-functionally with ML engineers, researchers, and product teams to align dataset structure with model training requirements, accelerating sprint delivery by 20%
- Engineered new features from raw tabular data through transformation, binning, and encoding techniques, boosting model performance.
- Automated ETL pipelines for tabular data ingestion and transformation using Pandas and SQL, reducing data preparation time by 50%
- Detected and handled outliers and skewed distributions across numerical features, improving training stability and reducing model error rates by 12-14%

China State Construction and Engineering Corporation Limited Multan, Pakistan

Business or Sector Construction | **Department** Intelligent Transportation Systems | **Website** https://csci.cscec.com/ en/xwzx/xw/201911/2992228.html

Intelligent Systems Engineer

08/10/2022 – 01/10/2024

- Deployed and maintained intelligent monitoring systems across 392 km of the Multan–Sukkur Motorway (M-5), ensuring 99.5% system uptime throughout the project lifecycle
- Integrated 50+ IoT sensors and smart devices along the motorway corridor for real-time traffic, structural, and environmental data collection, reducing manual inspection needs by 40%
- Built data dashboards and monitoring reports tracking KPIs across structural health, traffic density, and system performance for review by senior engineers and project directors
- Collaborated with cross-national engineering teams (Chinese and Pakistani engineers) across design, construction, and systems integration phases of a $2.9 billion infrastructure project
- Coordinated with telecom and hardware vendors to procure and install intelligent systems components, delivering 3 major milestones on schedule
- Configured automated alert systems for road anomalies and infrastructure faults, cutting incident response time by 35%

Full Grip Ecom Islamabad, Pakistan

Business or Sector Manufacturing | **Website** www.fullgripecom.com

Team Lead

- Led and managed a team of 8+ members across customer support, sales, and technical operations, maintaining 100% task completion rate on weekly deliverables
- Implemented performance tracking frameworks and KPI dashboards, improving overall team efficiency by 35% within the first 3 months
- Managed and resolved 200+ client queries per month across email, chat, and ticketing systems, maintaining a client satisfaction rate of 96%
- Reduced average query resolution time from 48 hours to under 6 hours by introducing priority-based ticket routing and response templates

- Integrated AI-powered chatbots into 10+ client Shopify stores, automating up to 65% of repetitive customer queries and reducing support workload significantly
- Deployed GPT-based conversational agents for order tracking, FAQs, and product recommendations, increasing customer engagement by 40% and upsell conversion by 18%
- Led the company-wide integration of AI tools (CRM automation, AI copywriting, predictive analytics) into internal operations, saving 25+ hours/week in manual work

Upwork Multan

Business or Sector Other service activities | **Website** https://www.upwork.com/freelancers/~01d4de3ea5f30fedf2

Freelance

11/04/2021 – Current

- Delivered 15+ generative AI projects across NLP, computer vision, and LLM integration, maintaining a 1**00 % Job Success Score** on Upwork
- Built and deployed RAG (Retrieval-Augmented Generation) pipelines for 5+ clients, reducing manual document search time by up to 70%
- **Fine-tuned LLMs (LLaMA, GPT, Mistral)** on domain-specific datasets, improving task accuracy by **25–40%** over base model baselines
- Developed AI-powered chatbots and virtual assistants for 8+ clients, cutting customer support response time by 60% and reducing support ticket volume by 35%
- Automated content generation workflows using GPT-4 and Claude APIs, saving clients an average of 20+ hours/week in manual content production
- Integrated vector databases (Pinecone, Weaviate, ChromaDB) into production systems, enabling semantic search across 1M+ document corpora

EDUCATION AND TRAINING

16/10/2025 – CURRENT

M.S. Computer Engineering Bahauddin Zakariya University, Multan

Website www.bzu.edu.pk | **Field of study** Engineering, manufacturing and construction | **Final grade** 3.97/4

30/06/2018 – 30/06/2022 Pakistan

BSc Computer Engineering (PEC # COMP/023391) University of Engineering and Technology, Taxila

Website web.uettaxila.edu.pk

05/03/2016 – 08/05/2018 Multan, Pakistan

FSC Pre-Engineering Punjab College Of Information And Technology, Multan

Address 60000, Multan, Pakistan

14/06/2015 – 16/07/2016 Multan, Pakistan

Matriculation Zakariya Public School, Multan

LANGUAGE SKILLS

MOTHER TONGUE(S): Urdu

Other language(s):

English

Listening C2
Spoken production C1
Reading C1
Spoken interaction C1
Writing C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

CERTIFICATIONS

CampusX, **09/02/2026**

Explainable AI(XAI)

Mode of learning: Online

Google Cloud, **08/02/2026**

Introduction to Large Language Models

Mode of learning: Online

Link coursera.org/verify/YE0SV79GTZUH

NVIDIA, **26/02/2026**

AI Infrastructure and Operations Fundamentals

Mode of learning: Online

Link https://www.coursera.org/account/accomplishments/verify/ZXTHQ28YSS0A

SKILLS

PROGRAMMING LANGUAGES

python | Java | C++ | MATLAB

MANAGEMENT SKILLS

Business Development Business Strategies | team leader / team player | Customer Services

COMPUTING TOOLS

OpenNN | Artificial Intelligence frameworks: Caffe, Darknet, YOLO | Deep Learning frameworks: PyTorch,Tensorflow,Keras | Google AutoML | Pytorch,Tensorflow | Eclipse, Microsoft Visual Studio, Visual Studio Code, NetBeans, IntelliJ | Linux (Ubuntu and Raspbian) | Raspberry Pi | Basic knowledge of Java C C++ C | Netbeans | Anaconda : Spyder/Jupyter | KALI LINUX | Microcontrollers | Arduino | Transformers

PERSONAL STRENGTHS

Research and development | Process & Performance improvement | Quality Assurance & Quality Control | Experience in management and leadership roles outside of work

PROJECTS

04/02/2026 – 10/03/2026

RAG PDF Chat — Conversational AI for Private Documents

Built and deployed a production-grade Retrieval-Augmented Generation application enabling context-aware conversations with private PDF documents without exposing sensitive data to third-party services. Engineered a multi-stage ingestion pipeline covering extraction, text normalization, and recursive chunking into a local FAISS vector store using HuggingFace sentence-transformers on CPU, ensuring zero data leakage. Implemented multi-step query refinement that rewrites ambiguous follow-ups into standalone retrieval queries before searching, improving answer accuracy on conversational threads. Integrated Groq's LLM API with an automatic four-model fallback chain eliminating single-point-of-failure on free-tier endpoints. Resolved non-trivial deployment challenges including Python 3.14 and Pydantic v1 incompatibility, gated model access restrictions, and serverless endpoint cold-start failures on Streamlit Cloud.
Github link -> https://github.com/zunairzafar/Conversationalapp

Link https://conversationalapp.streamlit.app/

04/06/2021 – 05/06/2022

Artificial Intelligence Based Affective Mirror

Artificial Intelligence based smart mirror designed in a home environment and to facilitate the users every day life, focusing on morning routines. The goal was to develop a Smart Mirror with a smooth and natural interaction behavior between users and The Smart Mirror, as well as investigate what content that was of interest for the users. As the technology is advancing and people are more inclined towards the AI and smart life, I decided to provide some contribution in order to ease life with the help of a AI based smart mirror. The UI was designed on the python script with the integration and parallelization of deep learning models for face detection and emotion recognition. With the help of a raspberry pi, the smart mirror then detect the face and then recognize the emotion based on the person's facial feature respectively. This type of algorithm was embedded with the UI to perform parallel operation at the same time.

Link https://www.youtube.com/watch?v=pnNxbVX1CyM&ab_channel=zunairzafar

01/12/2025 – 03/01/2026

Transformer Architecture from Scratch (PyTorch)

Designed and implemented a complete Transformer model from first principles using PyTorch, without relying on high-level library abstractions. The project includes manual implementation of multi-head self-attention, scaled dot-product attention, positional encoding, residual connections, layer normalization, and feed-forward networks. Emphasis was placed on tensor shape management, attention score computation, and end-to-end training stability. The model was trained and validated to demonstrate correct handling of contextual dependencies and sequence modeling, providing a deep, practical understanding of the core architecture underlying modern large language models.

Link https://github.com/zunairzafar/Pytorch/blob/main/Transfomers_from_Scratch.ipynb

13/01/2026 – 07/02/2026

Brain Tumor Classification

Developed an end-to-end brain tumor classification pipeline for MRI analysis, focusing on reproducibility and deployment constraints. The model, based on ResNet-18 and fine-tuned in PyTorch, was trained with strict preprocessing controls to ensure consistency. Model performance was evaluated using precision, recall, F1 scores, and confusion matrix analysis. Deployment challenges, such as packaging the model for cloud execution and managing GitHub size constraints, were addressed using Streamlit and GitHub Releases.
Streamlit -> https://resnet18.streamlit.app/

Link https://github.com/zunairzafar/Brain-Tumor-Classification

Gradient Descent Visualization

This project provides an interactive and animated visualization of the gradient descent optimization algorithm, developed using Streamlit. It allows users to observe how gradient descent iteratively adjusts parameters to minimize a cost function on a synthetic dataset, simulating a simple linear regression problem. Users can interact with controls to modify the learning rate, initial point, and number of iterations, offering insights into the algorithm's behavior. The visualization includes a dynamic 3D plot for Gradients,Loss function, showing the descent path and highlighting key concepts such as learning rate and convergence. Technologies like Streamlit, NumPy, Matplotlib, and Pandas are used to build the app and compute the necessary operations, making it an effective educational tool for understanding gradient descent.
Github: https://github.com/zunairzafar/Gradient_Descent_Visualization

Link https://gradientvisualization.streamlit.app/

Backpropagation Visualization

This project offers an interactive and animated visualization of the backpropagation algorithm, developed using Streamlit. It demonstrates how backpropagation is used to train neural networks by adjusting weights through gradient descent. The visualization shows the forward and backward passes, weight adjustments, and error minimization in real-time, enhancing understanding of key concepts like gradient flow and weight updates. Streamlit, NumPy, Matplotlib, and Pandas are employed to create the dynamic app and handle the calculations, making this tool a valuable resource for learning backpropagation in neural networks.
Github: https://github.com/zunairzafar/backpropagation

Link https://backpropagationn.streamlit.app/

Confidence Interval App for Numerical Analysis

This interactive Streamlit app visualizes confidence intervals, allowing users to explore how sample size, confidence level, and other parameters affect the interval. Users can adjust variables like sample size, population mean, and standard deviation to see real-time changes in the interval. The app displays dynamic visuals of data distributions and confidence intervals, helping users understand their role in statistical inference. Built with Streamlit, NumPy, Matplotlib, and Pandas, this tool provides an engaging way to learn about confidence intervals.

Link zunair-confidence-interval.streamlit.app